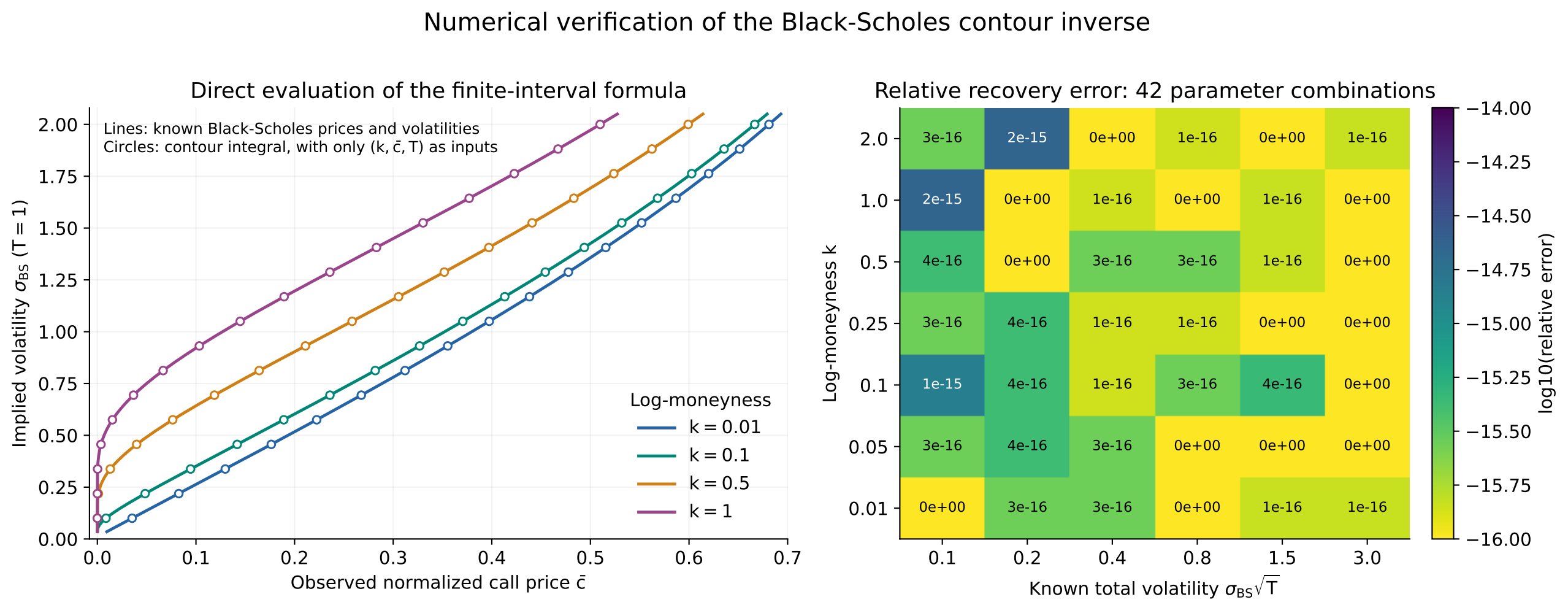
Numerical verification of the Black-Scholes contour inverse
Direct evaluation of the finite-interval formula
Lines: known Black-Scholes prices and volatilities
Circles: contour integral, with only (k, c̄, T) as inputs
Implied volatility σ_{BS} (T = 1)
Observed normalized call price c̄
Log-moneyness
k = 0.01
k = 0.1
k = 0.5
k = 1
Relative recovery error: 42 parameter combinations
Log-moneyness k
Known total volatility $\sigma_{BS}\sqrt{T}$
log10(relative error)
3e-16 2e-15 0e+00 1e-16 0e+00 1e-16
2e-15 0e+00 1e-16 0e+00 1e-16 0e+00
4e-16 0e+00 3e-16 3e-16 1e-16 0e+00
3e-16 4e-16 1e-16 1e-16 0e+00 0e+00
1e-15 4e-16 1e-16 3e-16 4e-16 0e+00
3e-16 4e-16 3e-16 0e+00 0e+00 0e+00
0e+00 3e-16 3e-16 0e+00 1e-16 1e-16